

AreaHustle Voice ASR Benchmark

Evaluation of intra-sentential Nigerian Pidgin-English code-switching for autonomous gig-marketplace transactions

Competition: Sahara CodeSwitch Africa Challenge, Track 5 **Date:** September 2026 **Dataset:** 20 samples, 2.8 min total **Version:** 1.0 (Final)

1. Executive summary

AreaHustle is a voice-first gig marketplace for Lagos informal markets. Customers post artisan jobs by speaking Nigerian Pidgin-English. This report evaluates three speech-to-text models on spontaneous Pidgin-English code-switched dispatch commands, then measures whether the resulting transcripts are accurate enough for an LLM agent to autonomously execute the downstream database transaction (trade category, location, budget) without human correction.

Intron Sahara v2.5 produces less than half the transcription error of the best global baseline (WER 26.6% vs 59.7%) and is the only model that preserves code-switched Pidgin markers and trade loanwords verbatim.

2. Models under evaluation

Model	Type	Parameters	Dialect	Inference
Intron Sahara v2.5	Domain-Specific API	Proprietary	pcm	Cloud API
OpenAI Whisper Large-v3	General Transformer	1.55B	English decoding	Colab T4 GPU
Meta MMS-1B	Multilingual CTC	1.0B	pcm	Colab T4 GPU

3. Dataset methodology

Property	Specification	Property	Specification
Source & Domain	Lagos informal artisan dispatch	Sample Count	20 consented recordings
Language / Code	Nigerian Pidgin (pcm)	Total Duration	2.8 minutes (avg 8.4 s / clip)
Audio Format	16 kHz, 16-bit mono WAV	Normalization	Peak-normalized (-1.0 dBFS)
Target Slots	Category, Location, Budget (NGN)	Ground Truth	Verbatim phonetic alignment

4. Evaluation metrics

- WER (Word Error Rate):** Standard benchmark for transcription fidelity, computed with `jiwer` after normalization (lowercase, punctuation stripped, whitespace collapsed).
- CER (Character Error Rate):** Particularly important for indigenous African terms, where minor character alterations separate intelligible slang from lexical corruption.
- Downstream Slot-Filling Accuracy:** Each model's transcripts were passed through the identical Gemini (`gemini-3.5-flash-lite`) extraction prompt and compared against ground-truth target slots. A sample counts as end-to-end success only when all three slots (category, location, budget) match.

5. Quantitative results

5.1 Acoustic performance (speech-to-text)

Model	WER (%)	CER (%)
Intron Sahara v2.5	26.6	12.9
OpenAI Whisper Large-v3	59.7	24.0
Meta MMS-1B	75.7	31.7

5.2 Downstream agentic task performance (Gemini slot-filling)

Source Transcript	Category (%)	Location (%)	Budget (%)	End-to-End (%)
Intron Sahara v2.5	55.0	40.0	95.0	15.0
OpenAI Whisper Large-v3	80.0	20.0	95.0	20.0
Meta MMS-1B	60.0	30.0	65.0	5.0

5.3 Analysis

Sahara produces less than half the transcription error of the best global baseline. The advantage is concentrated where AreaHustle needs it: code-switched Pidgin markers and vernacular trade loanwords.

Budget extraction is where acoustic error converts directly into money. Sahara and Whisper both land the escrow amount 95% of the time, while MMS's numeral corruption (3500 becomes "10005 hundred", 9000 becomes 1000, three clips returned a budget of 0) drops it to 65%.

Whisper posts the highest category accuracy (80% vs Sahara's 55%) despite more than double the WER. Its English-biased output matches the target label vocabulary ("carpenter", "electrician"), while Sahara's verbatim Pidgin trade terms ("roof repairer", "brick layer") fail exact string-match against the ground-truth labels. This is label aliasing, not acoustic failure, and it motivated the canonical category-mapping layer now in the production backend.

Location is the weakest slot for every model (Sahara 40%, MMS 30%, Whisper 20%). Yoruba-origin Lagos place names like Ojuelegba, Ilupeju, Ketu, and Agege get phonetically corrupted regardless of model. At n=20 each clip is five percentage points, so the Sahara-Whisper end-to-end gap (15% vs 20%) is within noise.

6. Qualitative failure-mode analysis

6.1 Per-model error profiles

Intron Sahara v2.5 preserved Pidgin grammatical particles ("wey", "fit", "wan") and vernacular trade terms ("rewinder", "vulcanizer"). It struggles on specialist loanwords and truncated numerals. Ground truth "I need rewinder wey go check my pumping machine, my budget na 6k" came back as "I need rwanda wey go check my pumping machine my budget na 6": "rewinder" became "rwanda" and the budget collapsed from ₦6,000 to ₦6.

OpenAI Whisper Large-v3 exhibits standard-English bias that causes phonetic hallucination over Pidgin markers. Ground truth "Abeg I dey Ikeja underbridge now, my brake dey sound, I need mechanic for 5k" came back as "ID Ikeza under bridge. My break day sound. I need mechanic for 5K." Every Pidgin marker snapped to its English homophone. On sample_11 the same bias produced full pseudo-Yoruba hallucination.

Meta MMS-1B systematically respells trade terms at the character level ("mekanik", "plonbar", "briklayer"), merges words into run-ons ("lekeifese" for "Lekki phase 1"), and corrupts numerals severely enough that three clips returned a budget of 0.

6.2 Case study: Language-code sensitivity (pcm vs yo)

To understand what the pcm language-code choice trades away, we ran sample_11 through Sahara twice: once with pcm, once with yo. Ground truth: "Welder dey this area? Iron gate hinge don cut for Ojuelegba, 4500 naira dey."

Sahara pcm "Welder dey dis area iron gate in don cut for Uju lego 4500 naira dey" - Trade term and budget preserved - Place name corrupted ("Ojuelegba" became "Uju lego")	Sahara yo "Well that day this area, I youngate in done court for ojú elégba o, 400 naira day" - Place name recovered in native orthography - Trade term and budget corrupted
---	---

The two language codes fail in complementary ways. A wrong budget silently misprices an escrow contract; a wrong neighbourhood only degrades matching. AreaHustle therefore defaults to pcm and exposes yo as an explicit user toggle.

7. Conclusion and production mitigations

Intron Sahara v2.5 halves the transcription error of the best global baseline (WER 26.6% vs 59.7%) and is the only model that preserves code-switched Pidgin markers and trade loanwords verbatim. The remaining failures concentrate in two places: specialist trade-label vocabulary and Yoruba-origin place names. Both are addressed in the production backend:

- A canonical category-mapping layer that maps verbatim Pidgin trade terms ("roof repairer", "brick layer") to the database schema ("Roofer", "Bricklayer") using fuzzy matching.
- A per-language (pcm/yo) toggle that lets users switch language codes when location accuracy matters more than budget precision.

Per-sample transcripts and full metrics are available in `benchmarks/results.json`.